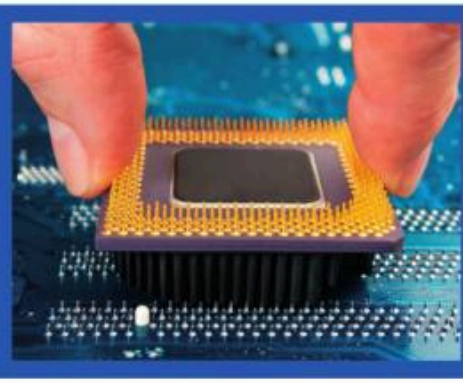
3 MEMORY AND PROCESSORS

Memory is used to store instructions. The processor executes those instructions. Both of these parts of the computer system have a big impact on the performance of the system. Specifically, they affect the speed at which software can be loaded and the speed at which software tasks can be completed



LEARNING OBJECTIVES

- Know that RAM stands for Random Access Memory and that ROM stands for Read Only Memory
- Be able to describe the characteristics of RAM and ROM, the differences between them and the impact on the user of the size of ROM/RAM
- Be able to describe the characteristics and uses of flash memory
- Understand the function of the processing unit (CPU)
- Know how the speed of a processor is measured

MEMORY

Memory can be accessed faster than secondary storage. Memory is used to store instructions so that the processor can fetch the instructions quickly in order to process them.

There are three types of memory that you need to know about:

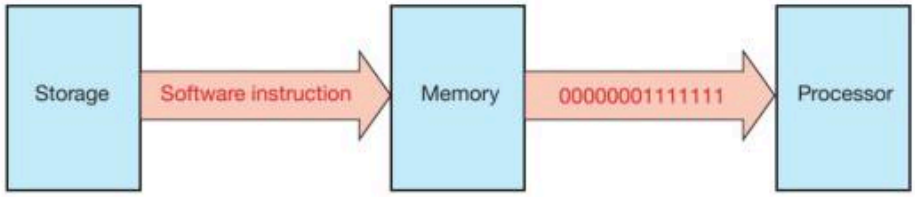
- Random Access Memory (RAM)
- Read Only Memory (ROM)
- Flash memory.

RANDOM ACCESS MEMORY (RAM)

HINT

- Two examples of secondary storage are hard disks and SSDs (see page 38).
- Memory is often called main memory or primary storage, and storage is often called secondary storage.

Software is loaded into RAM from a computer's secondary storage. The processor then fetches software instructions from memory and executes them.



▲ **Figure 3.1** Processors fetch software instructions from memory. The processor then decodes and executes the instructions

RAM is **volatile**, meaning that it cannot store data when it has no power. This means that, if you turn off your computer's power, any data held in RAM will be lost.

IMPACT OF THE SIZE OF RAM ON THE USER

RAM is used to store programs that are in use. The more RAM that is available, the more programs can run at the same time. This is important when you need to use many files or programs at the same time.

When the computer system does not have enough space in RAM, the operating system creates virtual memory by using an area of secondary storage. Secondary storage is much slower than RAM, so the user will notice a big decrease in system performance when the processor has to access instructions from virtual memory. When a solid state drive (SSD) is used as secondary storage, data is swapped to and from the SSD quickly and frequently. Although SSDs have faster access times than hard disk drives, this constant swapping can greatly reduce the performance of the SSD.

When programs or files are closed, they are removed from RAM. As space in RAM becomes available, active programs are moved into RAM from virtual memory, and this will improve the performance of those programs.

A user can add more RAM to a system, but only up to the maximum supported by the computer and operating system. Adding more RAM can improve the performance of a computer system when the user is using multiple files or programs at the same time. It can also be useful to increase the amount of RAM when programs are updated, as the updates often include more complex features that require the use of more memory.



▲ **Figure 3.2** A user can increase the amount of RAM using RAM slots on a motherboard

SKILLS INTELLECTUAL INTEREST AND CURIOSITY
REASONING

ACTIVITY

▼ USING RAM

1 Choose two operating systems. Research the maximum amount of RAM that can be installed for each one.

2 Research the cost of 4 GB of RAM today. What would that have cost last year? Investigate the price over the last five years and look at the price trend. What do you notice?

READ ONLY MEMORY (ROM)

ROM stores data permanently. Unlike RAM, ROM is **non-volatile**, meaning that data is not lost when the power is turned off. ROM is used in computer systems that store only one programme (single purpose computers), such as calculators, digital watches and washing machines. General purpose computers, such as home PCs and laptops, also use ROM to **boot** the system and load the operating system from secondary storage.

SUBJECT VOCABULARY

boot start up

- ROM is generally used to refer to memory that cannot be changed after manufacture. Its full name is Mask Programmed Read Only Memory.
- PROM stands for **Programmable** Read Only Memory. It is manufactured with the ability to be written to, but it can only be written to **once**.

Although the computer system cannot swap instructions in and out of ROM, it is possible to change the contents of some types of ROM so that the data on it can be updated. These types of ROM are called:

- EPROM (**Erasable Programmable** Read Only Memory)
- EEPROM (**Electrically Erasable Programmable** Read Only Memory).

The contents of EPROM and EEPROM can be erased and then rewritten to. Data stored in EPROM can be erased by exposing it to strong ultraviolet (UV) light. Data stored in EEPROM is erased by applying a voltage to one of the pins on the ROM chip. This means that you do not have to remove the ROM chip to erase the contents of EEPROM.

EEPROM later developed into flash memory (see pages 13 and 62 for more information about flash memory).

SUBJECT VOCABULARY

firmware a type of software that controls a hardware device

The process of erasing and rewriting the contents of EPROM or EEPROM is used when updating the **firmware** for a device (see page 57 for more information about software updates).

SKILLS INTELLECTUAL INTEREST AND CURIOSITY

ACTIVITY

▼ USING DIFFERENT TYPES OF ROM

- List five digital devices that use PROM that do not need to be edited.
- List five digital devices that use EPROM or EEPROM that can be edited.

DIFFERENCES BETWEEN RAM AND ROM

▼ **Table 3.1** Comparing the characteristics of RAM and ROM

	RAM	ROM (all types)
STORES DATA WHEN POWER OFF?	No (volatile)	Yes (non-volatile)
CAN SWAP INSTRUCTIONS IN AND OUT?	Yes	No
ACCESSIBLE IN ANY ORDER?	Yes	Yes
INTENDED TO STORE...	Temporary data	Permanent data
CAN BE UPGRADED?	Yes	No

FLASH MEMORY

Flash memory is a type of EEPROM. Like other types of ROM, it is **non-volatile**. It does not have any moving parts, so it has a fast access time and low power consumption. Because of its low power consumption, flash memory is used in the SSDs that are used in portable devices, such as laptop computers, as these devices often rely on internal batteries for their power. Flash memory is often used as removable storage in USB drives and SD cards.

PROCESSORS

A processor is made up of one or more Central Processing Units (CPUs). These carry out software instructions. In processors that are made up of more than one CPU, each CPU is referred to as a core. For example, in a quad-core processor, four cores are working during each **processor cycle**. This means that it can do up to four times as much work as a single-core processor.

SUBJECT VOCABULARY

processor cycle the process of fetching a program instruction from memory, decoding the actions required by the instruction and then executing those actions

DID YOU KNOW?

In multi-core processors, each core can run more slowly than in a single-core processor. This saves energy and produces less heat, which means that the processor requires less cooling. This makes the computer quieter as the fan does not have to cool the system. Reducing the amount of time in which the fan is running also reduces the amount of energy that the system uses, making the system more environmentally friendly and further increasing the battery life of mobile devices.

Processor speed is measured in **clock cycles per second**. This is the number of times per second the processor can carry out one or more instructions. Clock cycles are measured in units called hertz (Hz), kilohertz (kHz), megahertz (MHz) and gigahertz (GHz).

SUBJECT VOCABULARY

clock cycles per second used to measure processor speed; the number of times per second the processor can carry out one or more instructions

▼ Table 3.2 Measuring clock cycles

Name	Abbreviation	Clock cycles per second
Hertz	Hz	1
Kilohertz	kHz	1000
Megahertz	MHz	1000,000
Gigahertz	GHz	1000,000,000

Although clock cycles measure processing speed, there are other factors that affect how much work a processor can do in a given time. A user should also consider the amount of work that a CPU can do in each clock cycle. Different processors can carry out more instructions per core in each cycle.

SKILLS REASONING ADAPTIVE LEARNING

ACTIVITY

▼ WORKING AT DIFFERENT SPEEDS

It can be useful to think of processing speeds in relation to people or machinery working.

1 Two people are each moving a large pile of sand from one place to another by hand.
- Person A can move 100 portions every hour.
- Person B can move 10 portions every hour.

Who will finish first: Person A or Person B?

2 Now imagine that Person A uses a spoon to move each portion and Person B uses a digger to move each portion. Who will finish first now?

3 Now Person C is also trying to move their own pile of sand. They are using a digger at the same rate as Person B, but Person C's digger has four buckets (a 'quad-core digger'!). Who will finish first now?



▲ **Figure 3.3** The fastest does not always finish first

HINT

When thinking about the scenario in Question 2, consider that Person A moves more times per hour, but Person B moves more sand each time they move.

CHAPTER QUESTIONS

1 State what is meant by 'ROM'. **(1)**

2 State what is meant by 'RAM'. **(1)**

SKILLS REASONING

3 Explain why increasing the amount of RAM in a system improves the user experience. **(3)**

SKILLS INTERPRETATION

4 Describe what virtual memory is and why it is used. **(2)**

SKILLS INTERPRETATION

5 List **three** characteristics of RAM that make it different to ROM. **(3)**

SKILLS REASONING

6 Explain **two** reasons why the characteristics of flash memory make it suitable for use as secondary storage in smartphones. **(4)**

SKILLS INTERPRETATION

7 Which **one** of these describes the term 'clock cycle'? **(1)**

A The number of times per second that a processor can carry out instructions.

B The number of instructions that a processor can carry out each second.

C The number of times that a processor can load instructions for processing.

D The number of times that each process repeats in a processor.